

Informarl GNN代码阅读+运行 副本

代码运行指令

conda create -n infomarl --clone marl

pip uninstall onpolicy

[git@github.com:finleygou/InforMARL.git](https://github.com/finleygou/InforMARL.git)

pip install torch-geometric

pip install torch-scatter

pip install torch-sparse

添加setup.py

在onpolicy路径下添加__init__.py

导入global_var.py

```
(gnn_fom) goufandi@lthpc:/data/goufandi_space/Projects/GNN_rotational_formation/onpolicy$ cd ..
(gnn_fom) goufandi@lthpc:/data/goufandi_space/Projects/GNN_rotational_formation$ pip install -e .
Obtaining file:///data/goufandi_space/Projects/GNN_rotational_formation
  Preparing metadata (setup.py) ... done
Installing collected packages: onpolicy
  Running setup.py develop for onpolicy
Successfully installed onpolicy-0.1.0
(gnn_fom) goufandi@lthpc:/data/goufandi_space/Projects/GNN_rotational_formation$
```

让代码跑起来

修改路径

```
$ informarl_10n.sh M x global_var.py U out_25_1 train_mpe.py _init_.py U setup.py U
scripts > $ informarl_10n.sh
22 # cent_obs=("True" "False" "True" "False")
23 ep_lens=(25 50)
24
25 for seed in `seq ${seed_max}`;
26 do
27 # seed=`expr ${seed} + 1`
28 echo "seed: ${seed}"
29 # execute the script with different params
30 python ../onpolicy/scripts/train_mpe.py --use_valuenorm --use_popart \
31 --project_name "compare_${n_agents}" \
32 --env_name "GraphMPE" \
33 --algorithm_name "rmappo" \
34 --seed ${seed} \
35 --experiment_name "informarl_${ep_lens[$SLURM_ARRAY_TASK_ID]}" \
36 --scenario_name "navigation graph" \
```

更换wandb key

```
if all_args.use_wandb:
    # for supercloud when no internet_connection
    if not connected_to_internet():
        import json

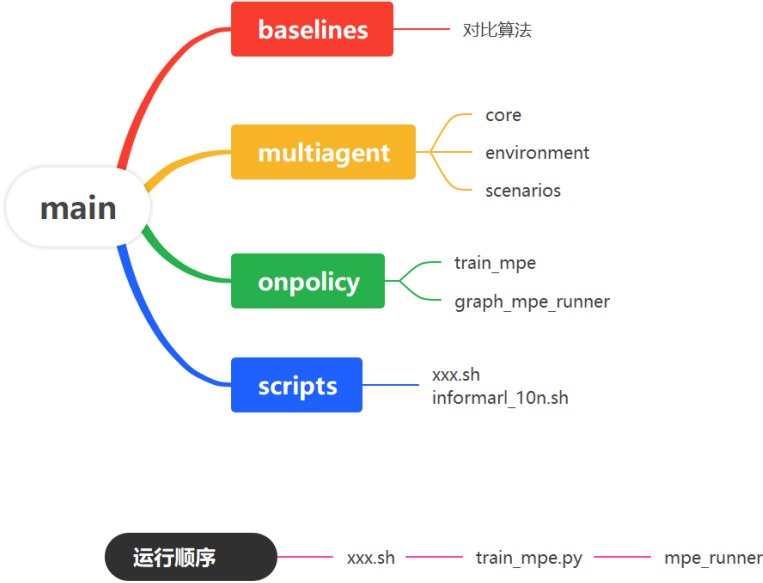
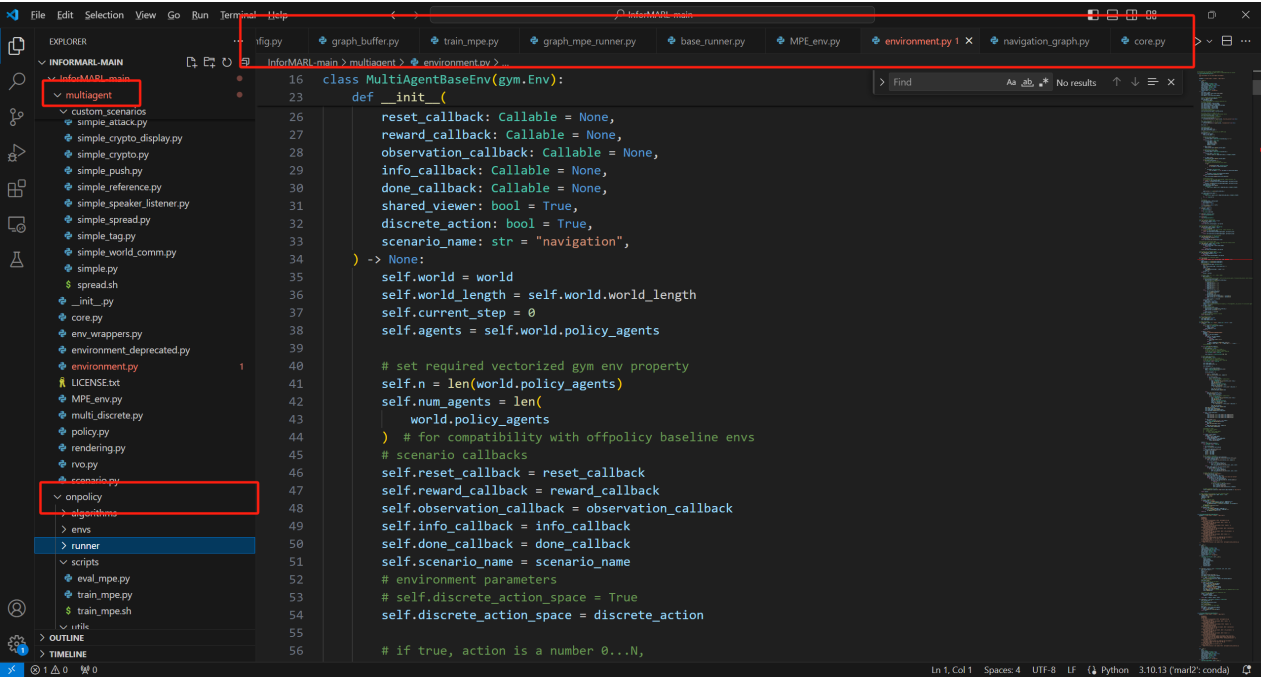
        # save a json file with your wandb api key in your
        # home folder as {'my_wandb_api_key': 'INSERT API HERE'}
        # NOTE this is only for running on systems without internet access
        # have to run `wandb sync wandb/run_name` to sync logs to wandboard
        os.environ["WANDB_API_KEY"] = "5aceaf53ea8aba2eca3b09e5b256acf7908e"
        os.environ["WANDB_MODE"] = "online"
        os.environ["WANDB_MODE"] = "dryrun"
        os.environ["WANDB_SAVE_CODE"] = "true"
```

将此处np.int改为np.int32

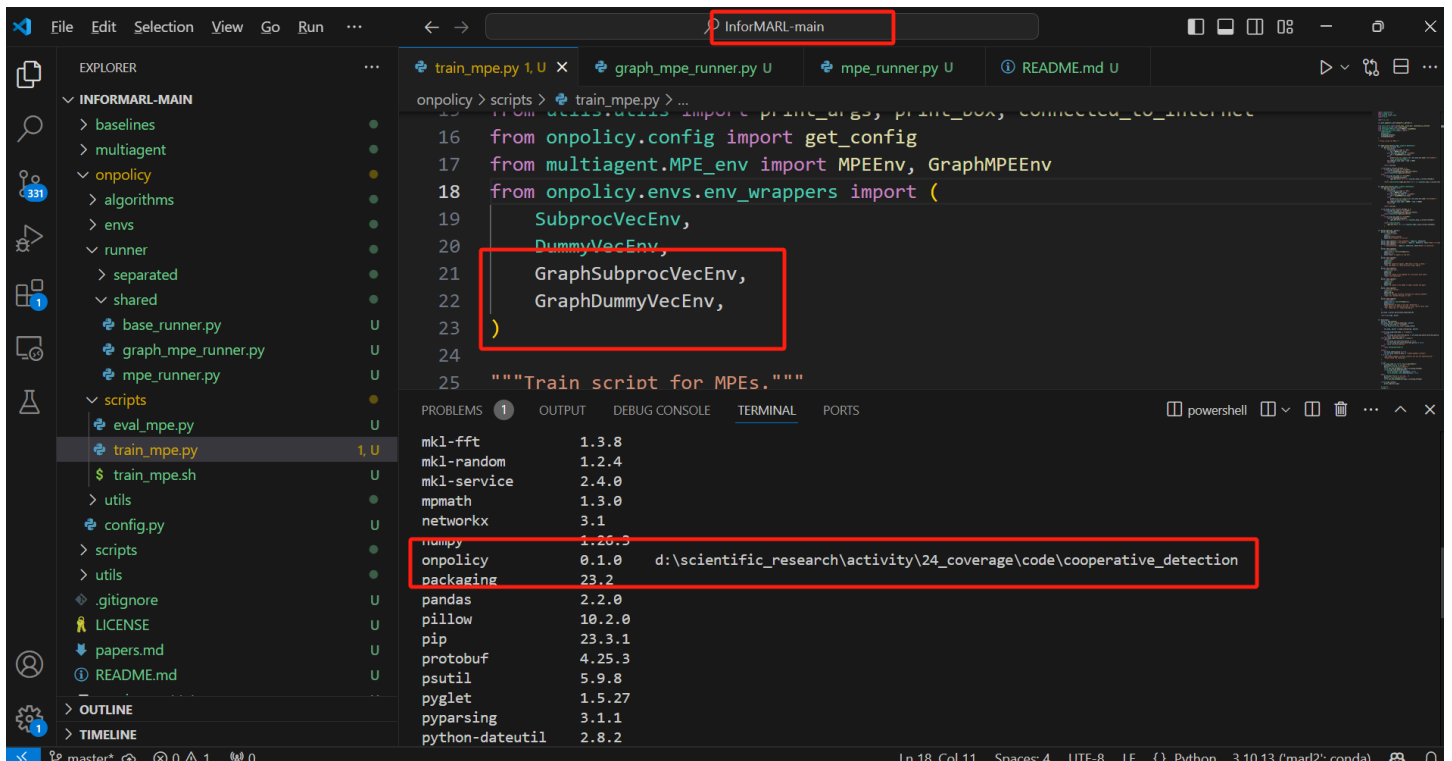
```
SEARCH np.int Aa _ab_*
Replace AB
8 results in 3 files - Open in editor
buffer.py baselines/dgn/dgn_navigation 1
...buffer_size, self.n_ant), dtype=np.int32)
graph_buffer.py onpolicy/utis 2
dtype=np.int32,
dtype=np.int32,
out_25_1 scripts/out_informarl10 5

onpolicy > utis > graph_buffer.py > GraphReplayBuffer > _init_
19 class GraphReplayBuffer(object):
45 def __init__(
117     num_agents,
118     *agent_id_shape,
119 ),
120 dtype=np.int32,
121 )
122 self.share_agent_id = np.zeros(
123 (
124     self.episode_length + 1,
125     self.n_rollout_threads,
126     num_agents,
127     *share_agent_id_shape,
128 ),
129 dtype=np.int32,
130 )
```

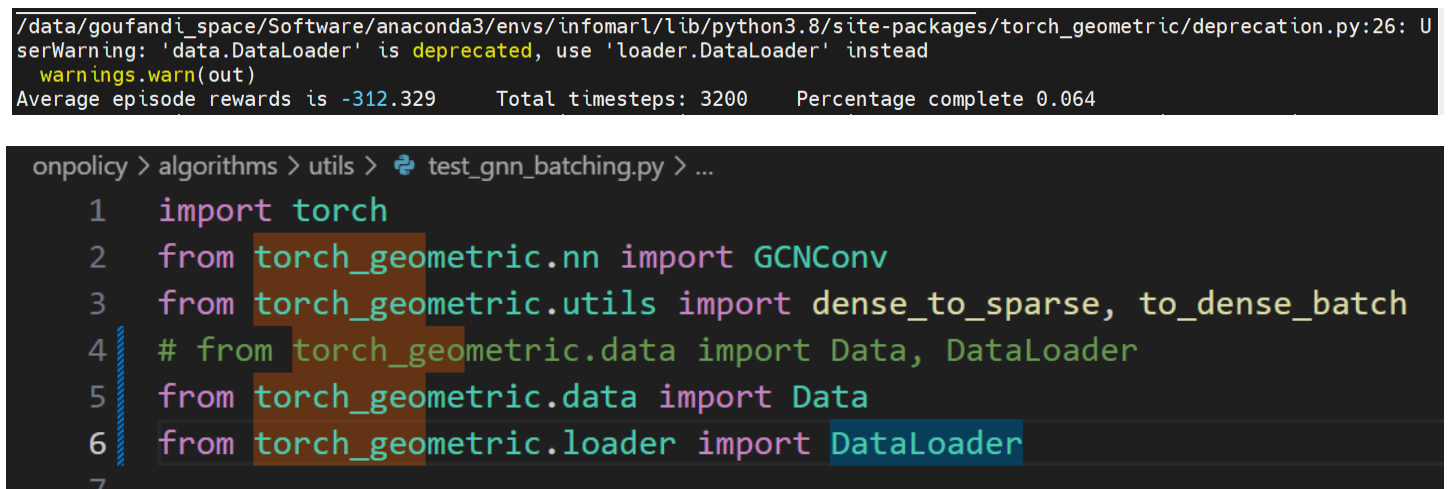
代码阅读顺序



之前的项目中, pip install -e.将onpolicy包下载到了相应的虚拟环境中,

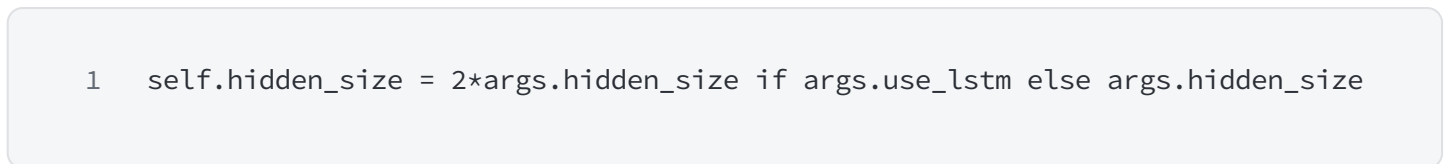


将data改成loader（利用vscode全局搜索将所有loader替换, gnn.py）



添加LSTM

将base_runner.py和graph_buffer.py中的hidden_size修改



这里是隐藏层的大小，rnn对应的就是hidden_size，lstm对应的是2*hidden_size(有h和c)。

添加了lstm.py，config里面新增了use_lstm参数，graph_actor_critic添加了LSTM layer。

添加globalvar

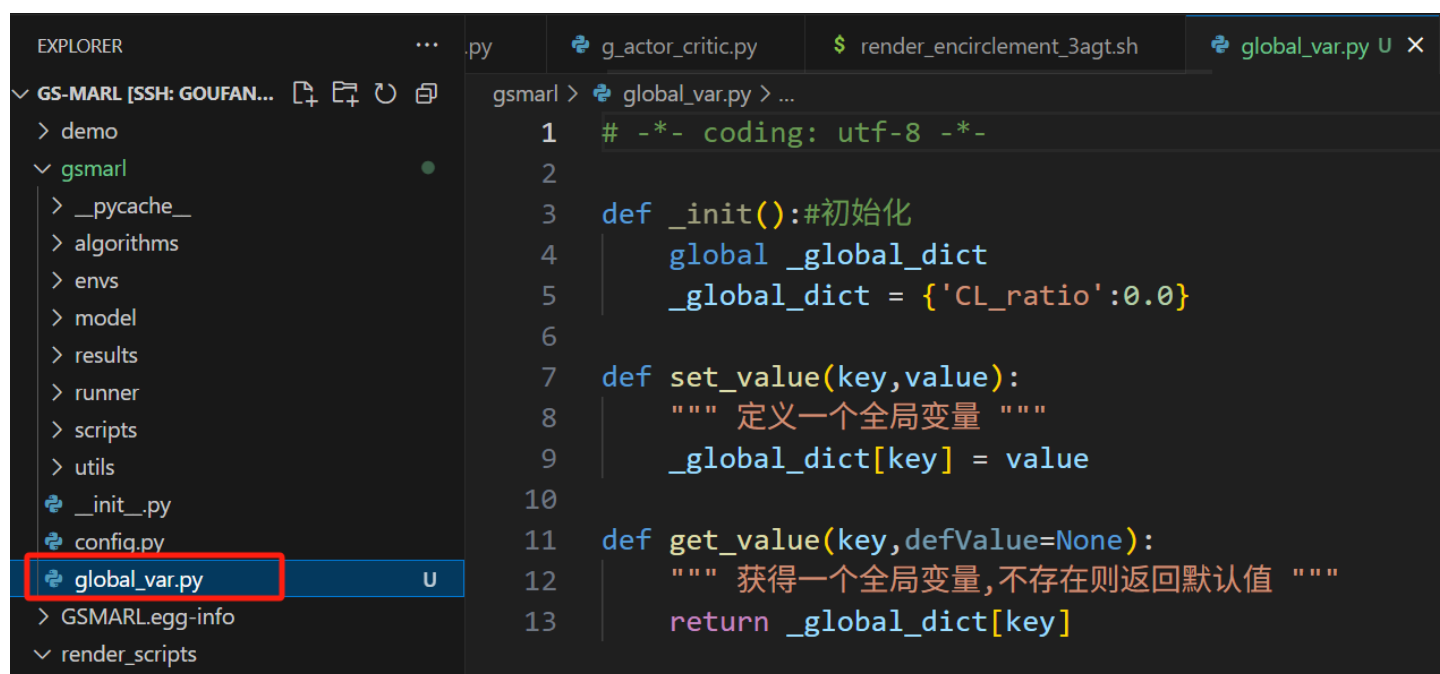
图相关

world.edge_list, 仅仅用来render

global_var.py, init.py, train_mpe.py, environment.py, simple_scenario.py

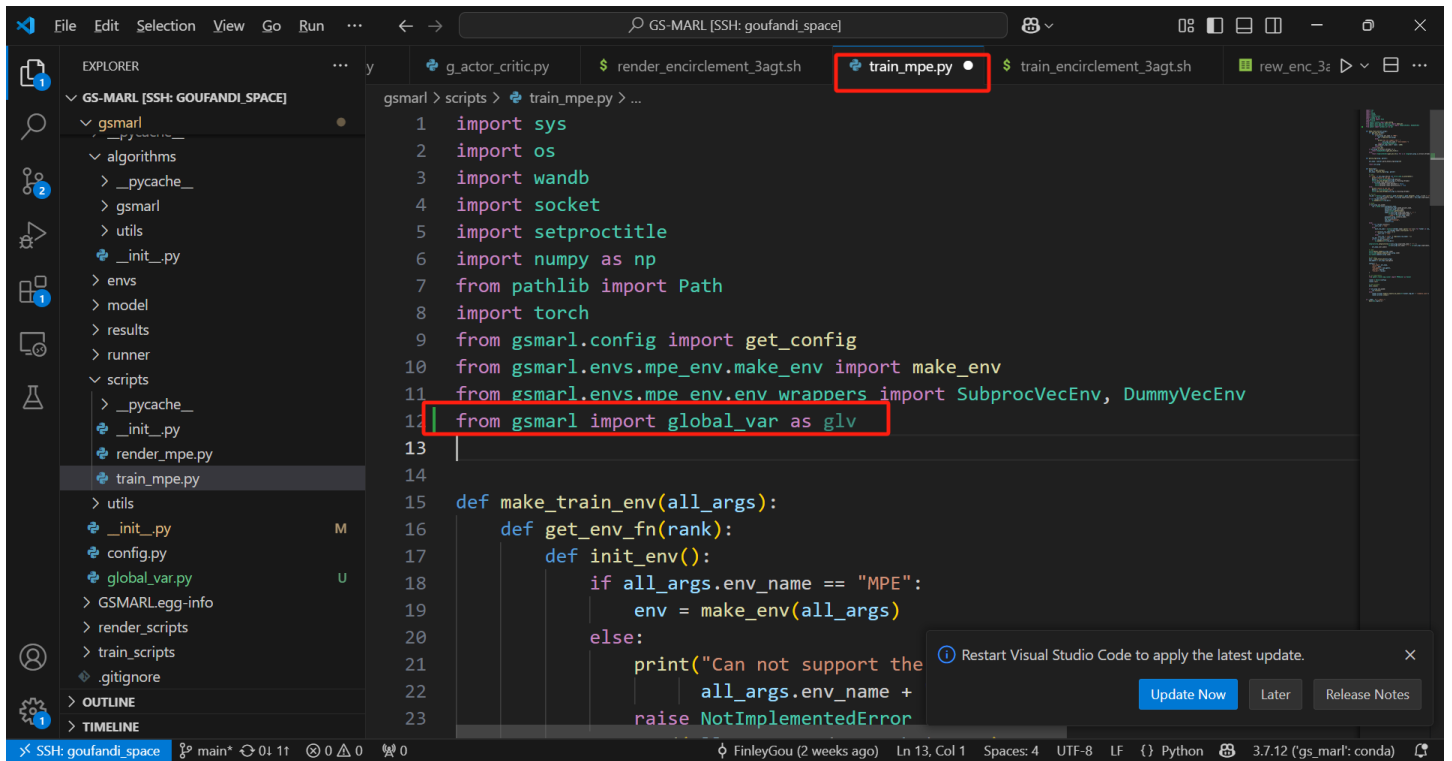
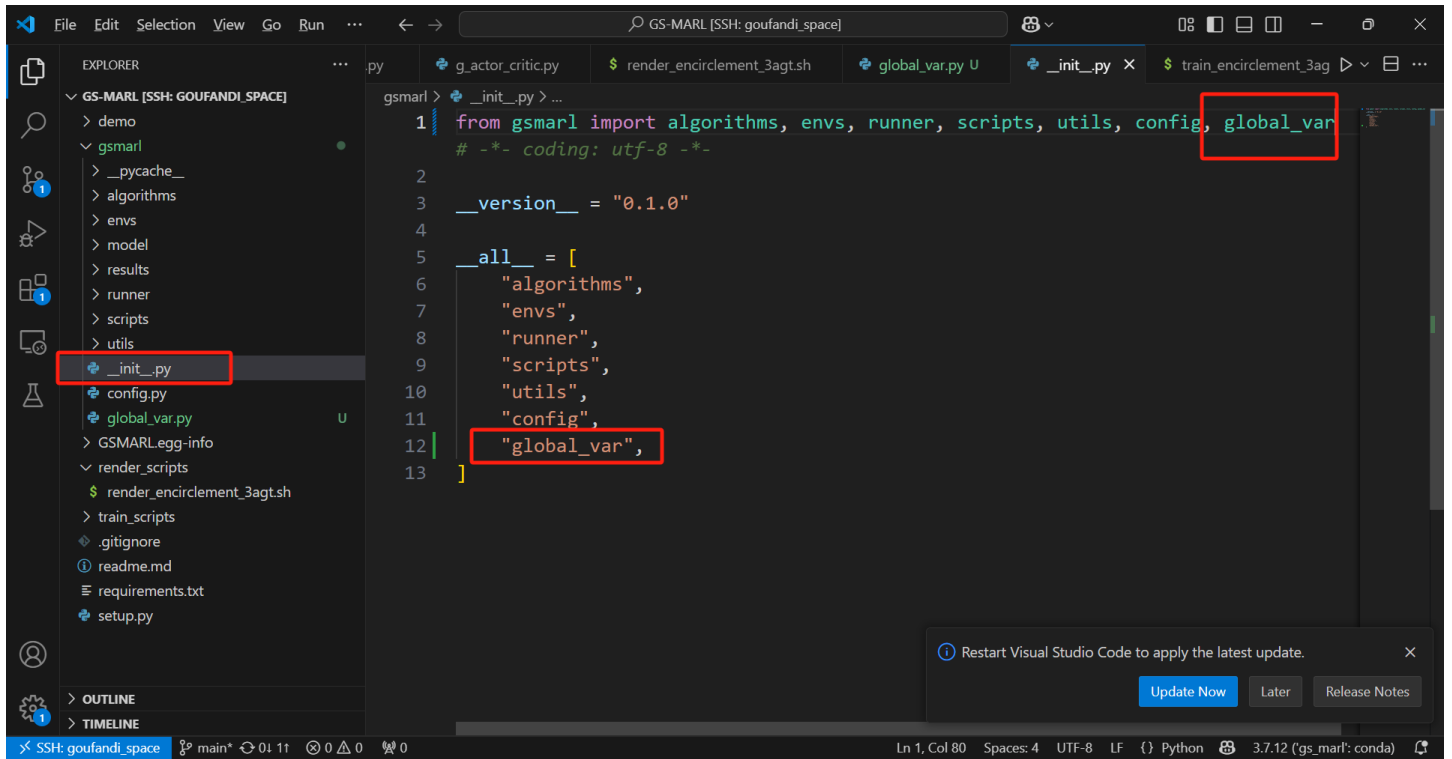
Envwrapper中Ctrl+F搜索_set_CL

区分 self.envs 和 self.remotes



The screenshot shows a VS Code editor interface. On the left, the Explorer sidebar displays a file tree for a project named 'GS-MARL [SSH: GOUFAN...]' with a subdirectory 'gsmarl'. Inside 'gsmarl', several files and folders are listed, including '__pycache__', 'algorithms', 'envs', 'model', 'results', 'runner', 'scripts', 'utils', '__init__.py', 'config.py', 'global_var.py' (which is selected and highlighted with a red box), 'GSMARL.egg-info', and 'render_scripts'. The main editor area shows the content of 'global_var.py'. The code includes a UTF-8 encoding declaration, an initialization function '_init()' that sets a global dictionary '_global_dict' with a 'CL_ratio' key, a 'set_value' function to update the dictionary, and a 'get_value' function to retrieve values from the dictionary, with a default value of None.

```
1  # -*- coding: utf-8 -*-
2
3  def _init():#初始化
4      global _global_dict
5      _global_dict = {'CL_ratio':0.0}
6
7  def set_value(key,value):
8      """ 定义一个全局变量 """
9      _global_dict[key] = value
10
11  def get_value(key,defValue=None):
12      """ 获得一个全局变量,不存在则返回默认值 """
13      return _global_dict[key]
```

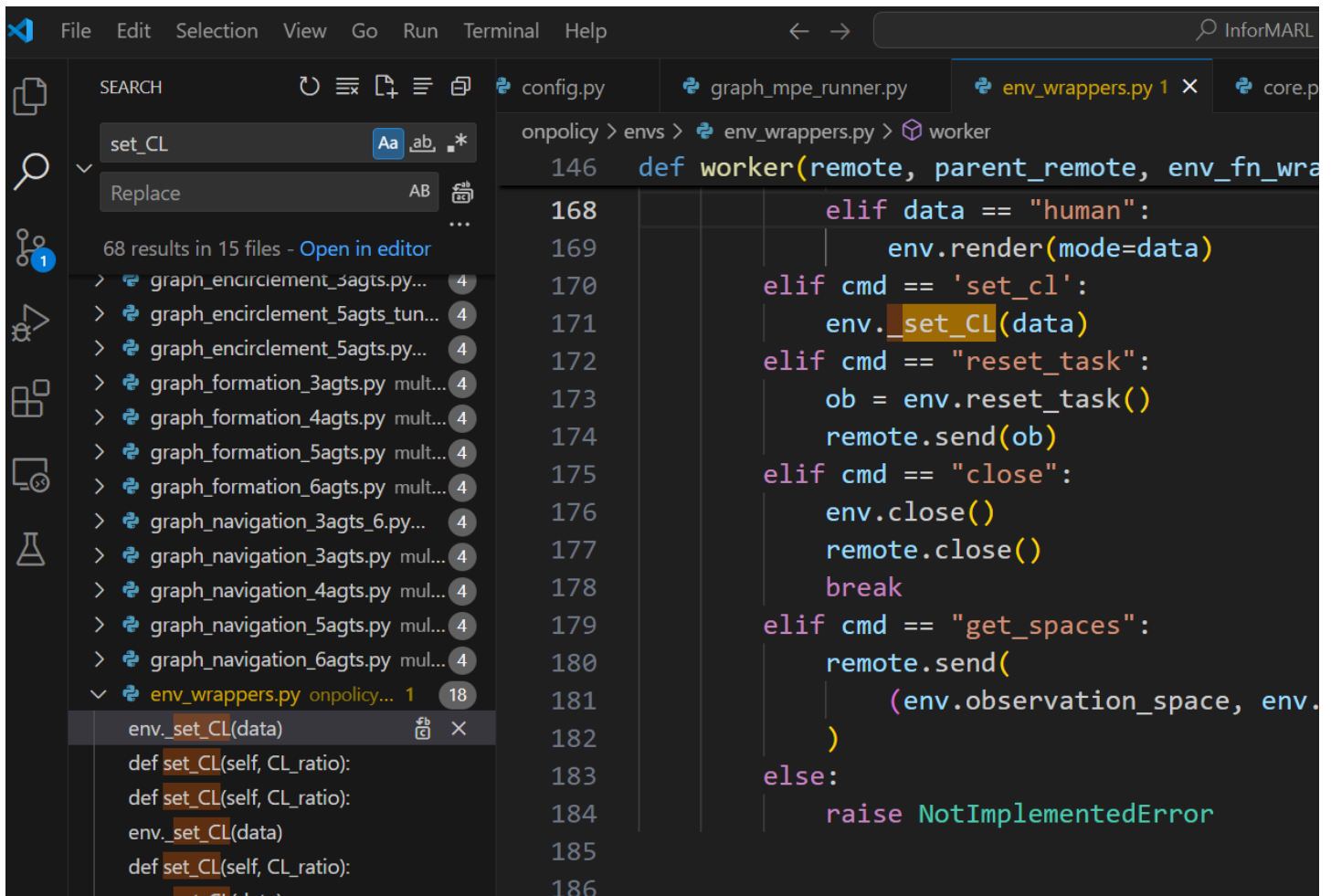


```
graph_mpe_runner.py X core.py $ train_encielement_5agts.sh $ train_navigation_5agts.sh
cy > runner > shared > graph_mpe_runner.py > GMPERunner > run

class GMPERunner(Runner):
    def run(self):
        # This is where the episodes are actually run.
        for episode in range(epochs):
            if self.use_linear_lr_decay:
                self.trainer.policy.lr_decay(episode, epochs)

            CL_ratio = episode/epochs
            glv.set_value('CL_ratio', CL_ratio) #curriculum learning
            self.envs.set_CL(glv.get_value('CL_ratio')) #env_wrapper
            for step in range(self.episode_length):
                # print("step:", step)
                # Sample actions
```

```
config.py graph_mpe_runner.py core.py $ train_encielement_5agts.sh $ train_navigation_5agts.sh train_mpe.py environment.py 1 X
multiagent > environment.py > MultiAgentBaseEnv
20 class MultiAgentBaseEnv(gym.Env):
230 def _set_action(self, action, policy_u, agent: Agent, action_space, time: Optional = None) -> None:
275 agent.action.u = policy_output
276 else:
277 act = network_output
278 agent.action.u = limit_action_inf_norm(act, 1)
279
280
281 def _set_CL(self, CL_ratio):
282 glv.set_value('CL_ratio', CL_ratio)
283 self.CL_ratio = glv.get_value('CL_ratio')
284
```



训练速度过慢

有效的手段：

增加rollout线程数

增加target_batch_size

split_batch=True # 理论上有效但是测试出来效果不佳

半精度训练（但是效果不佳，对收敛性有影响但是速度提升不够明显）

半精度训练

test1

--n_training_threads 16 --n_rollout_threads 128 \

--use_lstm "False" \

--episode_length \${ep_lens} \


```
--num_env_steps 5000000 \  
--ppo_epoch 15 --use_ReLU --gain 0.01 --lr 7e-4 --critic_lr 7e-4 \  
--user_name "finleygou" \  
--use_cent_obs "False" \  
--graph_feat_type "relative" \  
--use_att_gnn "True" \  
--split_batch "False" --max_batch_size 64 \  
--auto_mini_batch_size "True" --target_mini_batch_size 512  
done
```

<https://wandb.ai/finleygou/GraphMPE/runs/d2h68kgb>

FPS: 29

GPU占用率: 16% 460mb

test2 (没训完炸了)

```
--env_name "GraphMPE" \  
--algorithm_name "rmappo" \  
--seed ${seed} \  
--experiment_name "check" \  
--scenario_name "navigation_graph" \  
--use_wandb "True" \  
--num_agents=${n_agents} \  
--collision_rew 5 \  
--n_training_threads 16 --n_rollout_threads 128 \  
--use_lstm "False" \  
--episode_length ${ep_lens} \  
--num_env_steps 5000000 \  
--ppo_epoch 15 --use_ReLU --gain 0.01 --lr 7e-4 --critic_lr 7e-4 \  
--user_name "finleygou" \  
--use_cent_obs "False" \  
--graph_feat_type "relative" \  

```

```
--use_att_gnn "True" \  
--split_batch "False" --max_batch_size 64 \  
--auto_mini_batch_size "True" --target_mini_batch_size 2048  
done
```

target_batch_size = 2048

<https://wandb.ai/finleygou/GraphMPE/runs/nj8co45b>

FPS: 30

GPU占用率: 20% 654mb

9.19 test3

Rollout threads最多只能设128

```
--collision_rew 5 \  
--n_training_threads 16 --n_rollout_threads 128 \  
--use_lstm "False" \  
--episode_length ${ep_lens} \  
--num_env_steps 5000000 \  
--ppo_epoch 15 --use_ReLU --gain 0.01 --lr 7e-4 --critic_lr 7e-4 \  
--user_name "finleygou" \  
--use_cent_obs "False" \  
--graph_feat_type "relative" \  
--use_att_gnn "True" \  
--split_batch "True" --max_batch_size 2048 \  
--auto_mini_batch_size "True" --target_mini_batch_size 2048  
done
```

<https://wandb.ai/finleygou/GraphMPE/runs/motpgj37>

FPS: 28

三个实验的显存占用情况

						MIG M.	
0	NVIDIA GeForce RTX 3090	Off	00000000:31:00.0 Off	N/A			
34%	53C P2	116W / 350W	472MiB / 24576MiB	18%	Default	N/A	
1	NVIDIA GeForce RTX 3090	Off	00000000:4B:00.0 Off	N/A			
30%	45C P2	114W / 350W	670MiB / 24576MiB	19%	Default	N/A	
2	NVIDIA GeForce RTX 3090	Off	00000000:B1:00.0 Off	N/A			
33%	49C P2	123W / 350W	1793MiB / 24576MiB	20%	Default	N/A	
3	NVIDIA GeForce RTX 3090	Off	00000000:CA:00.0 Off	N/A			
30%	31C P8	21W / 350W	13MiB / 24576MiB	0%	Default	N/A	
Processes:							
GPU	GI ID	CI ID	PID	Type	Process name	GPU Memory Usage	
0	N/A	N/A	3790	G	/usr/lib/xorg/Xorg	4MiB	
0	N/A	N/A	2311235	C	rmappo-GraphMPE-check@finleygou	456MiB	
1	N/A	N/A	3790	G	/usr/lib/xorg/Xorg	4MiB	
1	N/A	N/A	2323786	C	rmappo-GraphMPE-check@finleygou	654MiB	
2	N/A	N/A	3790	G	/usr/lib/xorg/Xorg	4MiB	
2	N/A	N/A	2350070	C	rmappo-GraphMPE-check@finleygou	604MiB	
3	N/A	N/A	3790	G	/usr/lib/xorg/Xorg	4MiB	

(base) goufandi@lthpc:~\$